



University of Piraeus
School of Information and Communication Technologies
Department of Digital Systems

Level: Master's Programme of Studies

Cyber Defence and Digital Forensics

Assignment 2

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Exercise A – Volatility

- 1) In the memory image you will find information about the system's running processes (PIDs, PPIDs). Complete the table below with all visible processes.

Answer:

For the resolution of the first question and the analysis of the memory dump via the Volatility 3 tool, its installation on the Kali Linux system was required.

- The commands `sudo apt update` and `sudo apt install git python3 python3-pip python3-venv -y` were executed.
- Subsequently, to download the tool itself, the command `git clone https://github.com/volatilityfoundation/volatility3.git` was executed. **`https://github.com/volatilityfoundation/volatility3.git`**
- To avoid potential conflicts with already installed system libraries, it was deemed necessary to create an isolated virtual environment (venv).
- After activating the virtual environment, the installation of the necessary libraries for the correct operation of Volatility 3 followed.
- With the environment fully configured, we proceeded to analyse the memory file. To retrieve the list of running processes, the Volatility 3 `windows.pslist` plugin was used.

Analysing the results of the `windows.pslist` command, the `chrome.exe` process was identified with Process ID (PID) 4660.

kalaitzon@ubuntudesktop1: ~/volatility3										
PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	File output
4	0	System	0xfa80ccb9ae0 90	403	N/A	False	2020-07-22 09:02:18.000000 UTC	N/A	Disabled	
264	4	smss.exe	0xfa800dcf5480 2	29	N/A	False	2020-07-22 09:02:18.000000 UTC	N/A	Disabled	
352	340	csrss.exe	0xfa800e1d0060 9	486	0	False	2020-07-22 09:02:21.000000 UTC	N/A	Disabled	
404	340	wininit.exe	0xfa800eb37060 3	74	0	False	2020-07-22 09:02:24.000000 UTC	N/A	Disabled	
412	396	csrss.exe	0xfa800ebc3060 11	197	1	False	2020-07-22 09:02:24.000000 UTC	N/A	Disabled	
460	396	winlogon.exe	0xfa800ebbe060 4	111	1	False	2020-07-22 09:02:24.000000 UTC	N/A	Disabled	
504	404	services.exe	0xfa800ebcc260 10	210	0	False	2020-07-22 09:02:24.000000 UTC	N/A	Disabled	
512	404	lsass.exe	0xfa800ebd1b30 7	555	0	False	2020-07-22 09:02:24.000000 UTC	N/A	Disabled	
520	404	lsm.exe	0xfa800ebe5b30 11	146	0	False	2020-07-22 09:02:24.000000 UTC	N/A	Disabled	
624	504	svchost.exe	0xfa800ec52060 11	358	0	False	2020-07-22 09:02:24.000000 UTC	N/A	Disabled	
688	504	vmacthlp.exe	0xfa800ec8a060 4	53	0	False	2020-07-22 09:02:24.000000 UTC	N/A	Disabled	
720	504	svchost.exe	0xfa800ec9e470 7	275	0	False	2020-07-22 09:02:25.000000 UTC	N/A	Disabled	
820	504	svchost.exe	0xfa800ecd2b30 21	464	0	False	2020-07-22 09:02:25.000000 UTC	N/A	Disabled	
880	504	svchost.exe	0xfa800ed309e0 23	514	0	False	2020-07-22 09:02:27.000000 UTC	N/A	Disabled	
908	504	svchost.exe	0xfa800edc9b30 36	980	0	False	2020-07-22 09:02:27.000000 UTC	N/A	Disabled	
980	820	audiiodg.exe	0xfa800ed69330 7	129	0	False	2020-07-22 09:02:27.000000 UTC	N/A	Disabled	
416	504	svchost.exe	0xfa800eda2b30 12	518	0	False	2020-07-22 09:02:27.000000 UTC	N/A	Disabled	
568	504	svchost.exe	0xfa800ee07b30 21	380	0	False	2020-07-22 09:02:29.000000 UTC	N/A	Disabled	
1156	504	spoolsv.exe	0xfa800ee75670 13	265	0	False	2020-07-22 09:02:30.000000 UTC	N/A	Disabled	
1184	504	svchost.exe	0xfa800ede7b30 21	325	0	False	2020-07-22 09:02:30.000000 UTC	N/A	Disabled	
1364	504	VGAAuthService.	0xfa800eeeb30 4	86	0	False	2020-07-22 09:02:31.000000 UTC	N/A	Disabled	
1388	504	vmtoolsd.exe	0xfa800ef6a3a0 10	269	0	False	2020-07-22 09:02:31.000000 UTC	N/A	Disabled	
1836	624	WmiPrvSE.exe	0xfa800ef0f1060 10	197	0	False	2020-07-22 09:02:36.000000 UTC	N/A	Disabled	
1888	504	dllhost.exe	0xfa800ef0f0060 16	195	0	False	2020-07-22 09:02:37.000000 UTC	N/A	Disabled	
1980	504	msdtc.exe	0xfa800f163570 15	153	0	False	2020-07-22 09:02:38.000000 UTC	N/A	Disabled	
1648	624	WmiPrvSE.exe	0xfa800f2938b0 13	302	0	False	2020-07-22 09:02:56.000000 UTC	N/A	Disabled	
204	908	taskeng.exe	0xfa800f30d6a0 5	75	0	False	2020-07-22 09:03:10.000000 UTC	N/A	Disabled	
2364	504	svchost.exe	0xfa800f328060 10	140	0	False	2020-07-22 09:04:33.000000 UTC	N/A	Disabled	
2468	504	sppsvc.exe	0xfa800fc3e4c0 4	144	0	False	2020-07-22 09:04:34.000000 UTC	N/A	Disabled	
2536	2500	GoogleCrashHan	0xfa800fc8f7f0 6	97	0	True	2020-07-22 09:04:35.000000 UTC	N/A	Disabled	
2544	2500	GoogleCrashHan	0xfa800cea38e0 6	90	0	False	2020-07-22 09:04:35.000000 UTC	N/A	Disabled	
2556	504	svchost.exe	0xfa800ce6ab30 14	332	0	False	2020-07-22 09:04:35.000000 UTC	N/A	Disabled	
2688	504	SearchIndexer.	0xfa800ced1920 14	598	0	False	2020-07-22 09:04:35.000000 UTC	N/A	Disabled	
3004	504	WmiApSrv.exe	0xfa800f2de060 5	113	0	False	2020-07-22 09:05:02.000000 UTC	N/A	Disabled	
1624	504	taskhost.exe	0xfa800ceb2b30 10	156	1	False	2020-07-22 09:05:19.000000 UTC	N/A	Disabled	
1548	880	dwm.exe	0xfa800cf64b30 4	69	1	False	2020-07-22 09:05:19.000000 UTC	N/A	Disabled	
2188	596	explorer.exe	0xfa800d029b30 36	818	1	False	2020-07-22 09:05:19.000000 UTC	N/A	Disabled	
2428	2188	vmtoolsd.exe	0xfa800cfec710 9	196	1	False	2020-07-22 09:05:20.000000 UTC	N/A	Disabled	
1344	2188	chrome.exe	0xfa800cfffdb30 0	-	1	False	2020-07-22 09:06:37.000000 UTC	2020-07-22 09:07:03.000000 UTC	Disabled	
3000	880	WUDFHost.exe	0xfa800d1d8060 10	201	0	False	2020-07-22 09:07:29.000000 UTC	N/A	Disabled	
1008	2688	SearchProtocol	0xfa800d04d6a0 7	227	1	False	2020-07-22 09:07:38.000000 UTC	N/A	Disabled	
860	2688	SearchFilterHo	0xfa800cde3b30 5	81	0	False	2020-07-22 09:07:39.000000 UTC	N/A	Disabled	
1584	1388	cmd.exe	0xfa800e12fb30 0	-	0	False	2020-07-22 09:07:57.000000 UTC	2020-07-22 09:07:57.000000 UTC	Disabled	
3068	352	conhost.exe	0xfa800cf47b30 0	-	0	False	2020-07-22 09:07:57.000000 UTC	2020-07-22 09:07:57.000000 UTC	Disabled	
2916	1584	ipconfig.exe	0xfa800d06a8e0 0	-	0	False	2020-07-22 09:07:57.000000 UTC	2020-07-22 09:07:57.000000 UTC	Disabled	

Process ID (PID)	Parent Process ID (PPID)	Parent's Process ImageFileName	ImageFileName
1344	2188	explorer.exe	chrome.exe
2916	1584	cmd.exe	ipconfig.exe

2) What is the password of the user 'Adam'?

Answer:

In Windows operating systems, user passwords are never stored in plain text in memory — they are stored in the form of cryptographic hashes (NTLM or LM).

For the retrieval of the encrypted passwords, the windows.hashdump plugin of Volatility 3 was utilised. This specific plugin accesses the SAM (Security Accounts Manager) registry hive in memory.

Analysing the result of the above command, the entry corresponding to the user 'Adam' was located.

- **rid: 1000:** Constitutes the unique identifier of the user in the system.
- **lmhash: aad3b435b51404eeaad3b435b51404ee:** Contains a default, empty value indicating the system does not use LM hashing.
- **nthash: 7773c08920232397cae081704964b786:** This is the actual and useful hash of the password.

Consequently, the NTLM Hash of the user 'Adam' was determined as 7773c08920232397cae081704964b786.

To find the actual password in plaintext, the extracted NTLM hash was entered into the online platform crackstation.net.[CrackStation](https://crackstation.net)



CrackStation - Online Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:

7773c08920232397cae081704964b786

I'm not a robot

Crack Hashes

Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, rpeMD160, whirlpool, MySQL 4.1+ (sha1 sha1_bin), QubesV3.1BackupDefaults

Hash	Type	Result
7773c08920232397cae081704964b786	NTLM	qwerty123

Color Codes: Green Exact match, Yellow Partial match, Red Not found.

[Download CrackStation's Wordlist](#)

As confirmed by the platform's result, the hash was successfully identified. The password of the user 'Adam' is: Water1234!

3) How many times has the user 'Adam' executed the application 'Windows Remote Desktop'?

Answer:

To find the number of executions of the 'Windows Remote Desktop' application, the information of the UserAssist subsystem was utilised.

Since the UserAssist subsystem often records execution via the shortcut (.lnk) selected by the user, the exact path of the .lnk file was first identified.

Specifically, the part of the command 'python3 vol.py -f ~/Desktop/Investigation/windows.vmem' starts the Volatility framework.

Mapping the data of the rows with the table headers (Name | ID | Count | Focus Count | Time Focused):

ID: N/A

Count (Execution times): 8

Focus Count: 9

Therefore, based on the UserAssist data retrieved from memory, the integer number constituting the answer is 8.

- 4) When did the user 'Adam' open the file '1.jpg'?

Answer:

To determine the exact time at which the user 'Adam' opened the file '1.jpg', the investigation focused on the Windows registry subsystem known as 'RecentDocs'.

For reading and extracting this specific key directly from the RAM memory image, the windows.registry.hivelist plugin of Volatility 3 was utilised.

Analysing the command output, the entry relating to the user 'Adam's profile was successfully located.

- 5) When did the user search online for the phrase 'How to hack everything'?

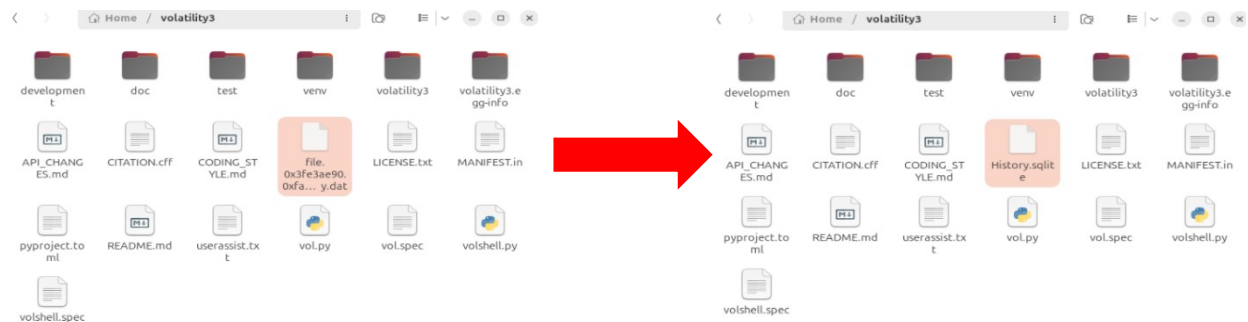
Answer:

For the resolution of the fifth question, it was necessary to locate and extract (dump) the browser's history database (History) from the memory image.

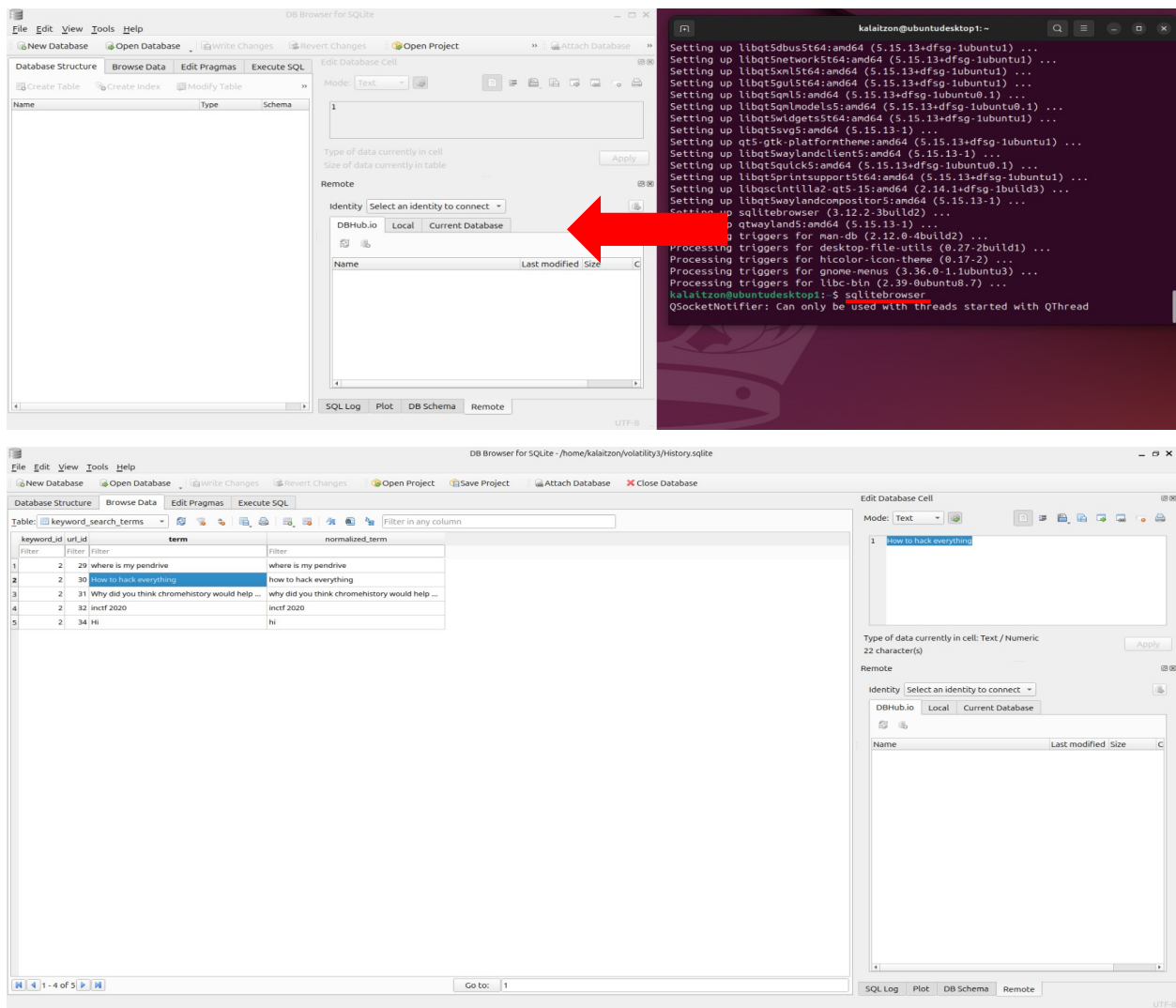
The first step involved finding the exact memory address (offset) at which the file was loaded. For this purpose, the windows.filescan plugin was used with a filter for the 'History' filename.

From the execution of the above command in the terminal, three different results (paths) emerged. From these, the path corresponding to the Chrome browser profile of the user Adam was isolated. For the extraction of the located file from the memory image, the 'windows.dumpfiles' plugin was selected.

To enable analysis of the extracted file's contents, it was deemed necessary to rename it with the appropriate extension (.sqlite).



Subsequently, the file was imported for further processing into the specialised application DB Browser for SQLite. Navigating to the tables of the database, the 'keyword_search_terms' table was selected.



During the navigation through the database data (Browse Data), the analysis initially focused on the keyword_search_terms table.

id	url	title	visit_count	typed_count	last_visit_time	hidden
4	https://www.microsoft.com/en-in/	Microsoft - Official Home Page	2	0	13239826606025199	0
5	https://twitter.com/_abhiramkumar	Abhiram Kumar (@_abhiramkumar) / Twitter	2	0	13239826334927477	0
6	https://twitter.com/_abhiramkumar	Abhiram Kumar (@_abhiramkumar) / Twitter	2	1	13239826334927477	0
7	https://github.com/stuxnet999	stuxnet999 (P. Abhiram Kumar) - GitHub	2	0	13239826400726233	0
8	https://github.com/stuxnet999	stuxnet999 (P. Abhiram Kumar) - GitHub	2	1	13239826400726233	0
9	https://www.bing.com/	Bing	2	2	13239826610168436	0
10	https://www.youtube.com/	YouTube	1	1	13239728206063724	0
11	https://www.linkedin.com/	LinkedIn: Log in or Sign Up	2	1	13239826422454455	0
12	https://chrome.com/	Google Chrome - Download the Fast, Secure ...	1	0	13239728259344933	0
13	https://chrome.com/	Google Chrome - Download the Fast, Secure ...	1	1	13239728259344933	0
14	https://www.google.com/chrome/	Google Chrome - Download the Fast, Secure ...	1	0	13239728259344933	0
15	https://nasa.gov/	NASA	1	1	13239728326415065	0
16	https://www.nasa.gov/	NASA	1	0	13239728326415065	0
17	https://mail.google.com/mail/tabrrm&ogbl	Gmail - Email from Google	1	0	13239728367901217	0
18	https://accounts.google.com/ServiceLogin?	Gmail - Email from Google	1	0	13239728367901217	0
19	https://mail.google.com/intl/en-GB/mail/help/...	Gmail - Email from Google	1	0	13239728367901217	0
20	https://www.google.com/intl/en-GB/mail/help/...	Gmail - Email from Google	1	0	13239728367901217	0
21	https://www.google.com/intl/en-GB/mail/help/...	Gmail - Email from Google	1	0	13239728367901217	0
22	https://outlook.com/	Outlook - free personal email and calendar ...	1	0	13239728375952334	0
23	https://outlook.com/	Outlook - free personal email and calendar ...	1	1	13239728375952334	0
24	https://www.outlook.com/owa/	Outlook - free personal email and calendar ...	1	0	13239728375952334	0
25	https://outlook.live.com/owa/	Outlook - free personal email and calendar ...	1	0	13239728375952334	0
26	https://nvidia.com/	Artificial Intelligence Computing Leadership ...	1	1	13239728382800605	0
27	https://www.nvidia.com/	Artificial Intelligence Computing Leadership ...	1	0	13239728382800605	0
28	https://www.nvidia.com/en-in/	Artificial Intelligence Computing Leadership ...	1	0	13239728382800605	0
29	https://www.google.com/search?...	where is my pendrive - Google Search	2	0	13239728473355260	0
30	https://www.google.com/search?...	How to hack everything - Google Search	2	0	13239826440787747	0
31	https://www.google.com/search?...	Why did you think chrome history would help ...	2	0	13239826470540741	0
32	https://www.google.com/search?...	Inc47 2020 - Google Search	2	0	13239826499124744	0

The extracted value (13239826440787747) constitutes a timestamp in WebKit format. This specific formatting is used natively by the Chrome browser.

Therefore, the final answer for question 5 is 2020-07-21 17:34:00 UTC.

Epilogue

Making a quick review of what we achieved with Volatility 3, we have now compiled a complete evidence folder of the digital investigation.

Exercise B – EvtxECmd

Before answering the individual questions, it was deemed necessary to prepare the set of Windows Event log files (.evtx) for analysis using the EvtxECmd tool.

- **-d Logs:** This parameter (-d for directory) indicated to the tool the directory ('Logs') within which the .evtx log files to be processed were located.
- **--csv .:** Gave the instruction for exporting the processed data in comma-separated (CSV) file format.

- **--csvf ParsedLogs.csv:** Specified the exact and final naming of the generated file as **ParsedLogs.csv**.

After completing the command execution, the tool successfully generated the consolidated file ParsedLogs.csv in the current directory.

```
C:\Windows\System32\cmd.e x + v
7026      1
7040      7
10016     2
16962     1
16977     1
16983     1
50036     1
50037     2
50103     1
50104     2
50105     2
50106     2
51046     1
51047     2
51057     2

Processing C:\Users\Kalai\Desktop\ExerciseB\Logs\Windows PowerShell.evtx...
Chunk count: 1, Iterating records...

Event log details
Flags: None
Chunk count: 1
Stored/Calculated CRC: 19838BEC/19838BEC
Earliest timestamp: 2024-10-29 22:13:28.0543565
Latest timestamp: 2024-12-24 22:50:47.1505326
Total event log records found: 16

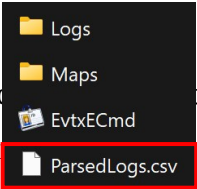
Records included: 16 Errors: 0 Events dropped: 0

Metrics (including dropped events)
Event ID      Count
400           2
403           2
600          12

Processed 110 files in 13.0864 seconds

Files with errors
C:\Users\Kalai\Desktop\ExerciseB\Logs\Microsoft-Windows-CertificateServicesClient-Lifecycle-System%4Operational.evtx error count: 1

C:\Users\Kalai\Desktop\ExerciseB>
```



Given the enormous volume of records generated, for effective overview, filtering and searching, the specialised Timeline Explorer tool is used.

Timeline Explorer v2026.5.0

File Tools Tabs View Help

ParsedLogs.csv

Drag a column header here to group by that column

Enter text to search... Find

Line	Tag	Record Number	Event Record Id	Time Created	Event Id	Level	Provider	Channel	Process
1	☐	957	957	2024-12-24 23...	6000	Info	Microsoft-Windows-Winlogon	Application	
2	☐	958	958	2024-12-24 23...	6000	Info	Microsoft-Windows-Winlogon	Application	
3	☐	959	959	2024-12-24 23...	1532	Info	Microsoft-Windows-User Profiles Service	Application	
4	☐	960	960	2024-12-27 00...	1531	Info	Microsoft-Windows-User Profiles Service	Application	
5	☐	961	961	2024-12-27 00...	100	Info	Bonjour Service	Application	
6	☐	962	962	2024-12-27 00...	100	Info	Bonjour Service	Application	
7	☐	963	963	2024-12-27 00...	100	Info	Bonjour Service	Application	
8	☐	964	964	2024-12-27 00...	5615	Info	Microsoft-Windows-WMI	Application	
9	☐	965	965	2024-12-27 00...	6003	Info	Microsoft-Windows-Winlogon	Application	
10	☐	966	966	2024-12-27 00...	6000	Info	Microsoft-Windows-Winlogon	Application	
11	☐	967	967	2024-12-27 00...	900	Info	Microsoft-Windows-Security-SPP	Application	
12	☐	968	968	2024-12-27 00...	16394	Info	Microsoft-Windows-Security-SPP	Application	
13	☐	969	969	2024-12-27 00...	1066	Info	Microsoft-Windows-Security-SPP	Application	
14	☐	970	970	2024-12-27 00...	1003	Info	Microsoft-Windows-Security-SPP	Application	
15	☐	971	971	2024-12-27 00...	902	LogAlways	Microsoft-Windows-Security-SPP	Application	
16	☐	972	972	2024-12-27 00...	5617	Info	Microsoft-Windows-WMI	Application	
17	☐	973	973	2024-12-27 00...	1003	Info	Microsoft-Windows-Security-SPP	Application	
18	☐	974	974	2024-12-27 00...	1003	Info	Microsoft-Windows-Security-SPP	Application	
19	☐	975	975	2024-12-27 00...	1003	Info	Microsoft-Windows-Security-SPP	Application	
20	☐	976	976	2024-12-27 00...	1003	Info	Microsoft-Windows-Security-SPP	Application	
21	☐	977	977	2024-12-27 00...	8198	Error	Microsoft-Windows-Security-SPP	Application	
22	☐	978	978	2024-12-27 00...	1003	Info	Microsoft-Windows-Security-SPP	Application	
23	☐	979	979	2024-12-27 00...	8198	Error	Microsoft-Windows-Security-SPP	Application	
24	☐	980	980	2024-12-27 00...	1003	Info	Microsoft-Windows-Security-SPP	Application	
25	☐	981	981	2024-12-27 00...	8198	Error	Microsoft-Windows-Security-SPP	Application	
26	☐	982	982	2024-12-27 00...	102	Info	ESENT	Application	

C:\Users\Kalan\Desktop\ExerciseB\ParsedLogs.csv

Total lines 35,254 Visible lines 35,254 Open files: 1 Search options

1) What is the recorded OS start time and OS shutdown time of the system?

Answer:

To find the recorded OS start time and OS shutdown time, the specialised Windows Event log entries were analysed.

During the initial search based exclusively on Event ID 12, the tool returned a multitude of results. Analysing them, it was found that this is because multiple Event Log services are recorded with the same Event ID.

Timeline Explorer v2026.5.0

File Tools Tabs View Help

ParsedLogs.csv

Drag a column header here to group by that column

Enter text to search... Find

Line	Tag	Record Number	Event Record Id	Time Created	Event...	Level	Provider	Channel	Pro
12803		23	23	2024-10-29 22:19:47	12	Info	Microsoft-Windows-Diagnosis-Scheduled	Microsoft-Windows-Diagno...	
16528		7	7	2024-04-01 09:39:37	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16529		8	8	2024-04-01 09:39:37	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16530		9	9	2024-04-01 09:39:41	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16531		10	10	2024-04-01 09:39:41	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16532		11	11	2024-04-01 09:39:50	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16533		12	12	2024-04-01 09:39:50	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16534		13	13	2024-04-01 09:40:20	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16535		14	14	2024-04-01 09:40:20	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16536		15	15	2024-04-01 09:41:53	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16537		16	16	2024-04-01 09:41:53	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16538		17	17	2024-04-01 09:45:37	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16539		18	18	2024-04-01 09:45:37	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16540		19	19	2024-04-01 09:57:00	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16541		20	20	2024-04-01 09:57:00	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16542		21	21	2024-04-01 09:57:03	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16543		22	22	2024-04-01 09:57:03	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16544		23	23	2024-04-01 09:57:11	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16545		24	24	2024-04-01 09:57:11	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16546		25	25	2024-04-01 09:57:40	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16547		26	26	2024-04-01 09:57:40	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16548		27	27	2024-04-01 09:59:03	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16549		28	28	2024-04-01 09:59:03	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	
16550		29	29	2024-04-01 09:59:55	12	Warning	Microsoft-Windows-Security-Mitigations	Microsoft-Windows-Securi...	

Event Id = 12

C:\Users\Kalah\Desktop\ExerciseB\ParsedLogs.csv

Total lines 35,254 Visible lines 200 Open files: 1 Search options

For the precise isolation of the actual boot and shutdown events, combined filtering was applied.

Timeline Explorer v2026.5.0

File Tools Tabs View Help

ParsedLogs.csv

Drag a column header here to group by that column

Enter text to search... Find

Line	Tag	Record Number	Event Record Id	Time Created	Event...	Level	Provider	Channel	Pro
35140		1456	1456	2024-12-27 00:04:51	12	Info	Microsoft-Windows-Kernel-General	System	

Event Id = 12 And Provider = Microsoft-Windows-Kernel-General

C:\Users\Kalah\Desktop\ExerciseB\ParsedLogs.csv

Total lines 35,254 Visible lines 1 Open files: 1 Search options

Timeline Explorer v2026.5.0

File Tools Tabs View Help

ParsedLogs.csv

Drag a column header here to group by that column

Enter text to search... Find

Line	Tag	Record Number	Event Record Id	Time Created	Event...	Level	Provider	Channel	Pro
35139		1455	1455	2024-12-24 23:14:09	13	Info	Microsoft-Windows-Kernel-General	System	

Event Id = 13 And Provider = Microsoft-Windows-Kernel-General

Total lines 35,254 Visible lines 1 Open files: 1 Search options

Since the EvtxECmd tool by default normalises all timestamps to the UTC time zone during export:

- **OS Shutdown (Event ID 13): 2024-12-24 23:14:09 UTC**
- **OS Start (Event ID 12): 2024-12-27 00:04:51 UTC**

It is worth noting that the chronological sequence is perfectly consistent, indicating that the system terminated its operation on 24 December and restarted on 27 December 2024.

- 2) What is the path of the file in question, what is its name, and what is the date it is supposed to have been created?

Answer:

This specific question concerns the identification of a classic anti-forensics technique, whereby a malicious file is hidden inside a legitimate file using Alternate Data Streams (ADS).

- **File path: C:\Users\phorger\Documents\desktop.ini**
- **Hidden file name (ADS): desktop.ini:vault.vhdx**

- **Creation date and time (UTC): 2024-12-24 22:57:32**

3) Identify from which log files the user in question cleared the entries, and when these actions took place.

Answer:

This question concerns one of the most critical indicators of system compromise: the deliberate clearing of event log files.

Initially, the search in the Timeline Explorer environment focused on Event ID 104. Filtering the results, it was found that the user phorger cleared the entries of the System, Application, and Security logs.

Since the security log file is the primary target during evidence concealment (as it records authentication events and privilege escalations), this entry was examined in detail.

In conclusion, the deletion of all three main log files within a minimum time window of just four seconds constitutes a clear indication of deliberate anti-forensics activity.

- **Application log: 2024-12-24 23:13:40**
- **System log: 2024-12-24 23:13:41**
- **Security log: 2024-12-24 23:13:44**

4) When did the administrator-level connection for the user 'phorger' take place, according to the log files?

Answer:

To determine the exact time at which the user phorger logged on with administrator privileges, the investigation proceeded with a targeted search of the parsed log files.

On Windows systems, a typical, successful logon is recorded with Event ID 4624 (Successful Logon). However, when special privileges are assigned to the user, Event ID 4672 (Special privileges assigned) is additionally recorded.

The methodology in Timeline Explorer involved combined filtering. Specifically, the filter 4672 was set in the 'Event Id' column.

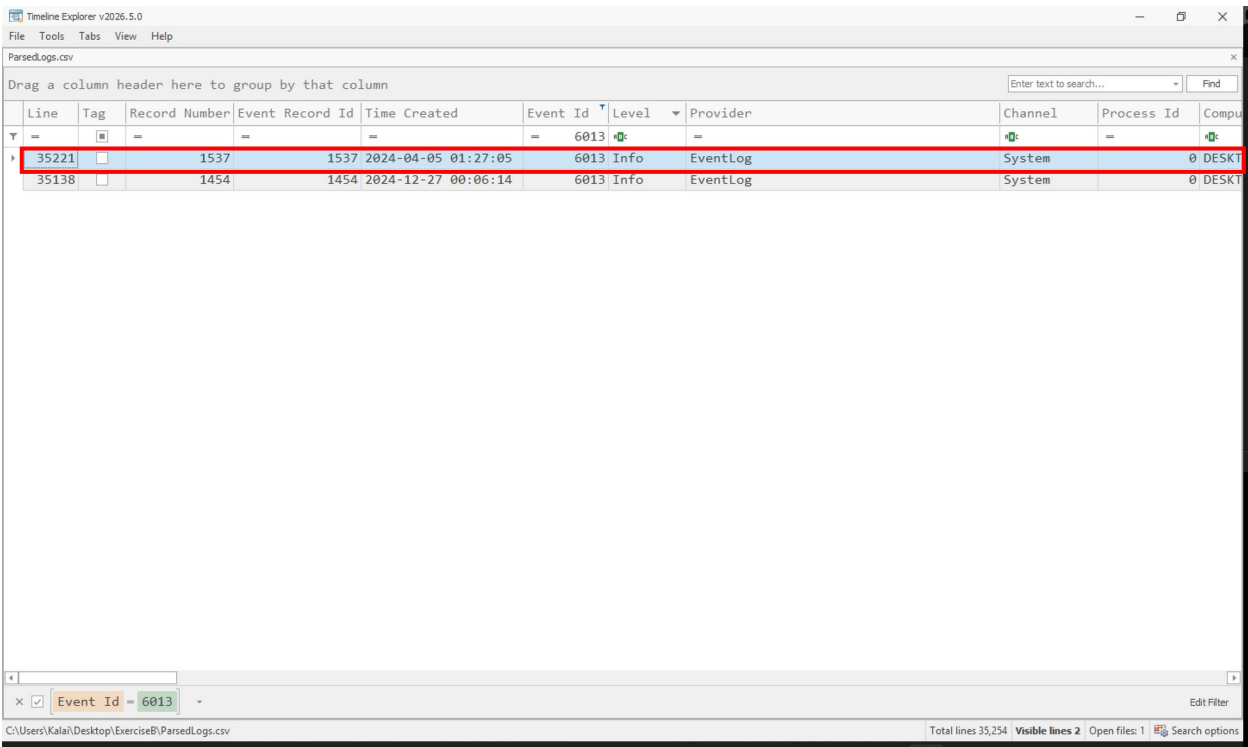
Therefore, based on the Security log file, the administrator-level connection for the user phorger took place at: 2024-12-24 22:56:00 UTC.

5) Based on the log files, what was the operating time (in seconds) of the system on '2024-04-05'?

Answer:

For the precise determination of the system's operating time on that specific date, it is not necessary to manually calculate the difference between start and shutdown events.

To retrieve this information in the Timeline Explorer environment, a new filter was applied in the 'Event Id' column, setting the value 12.



Examining the data of this specific record in the 'Payload Data1' column, the exact information generated by the Windows Kernel was retrieved.

Epilogue

The analysis of the log files revealed a complete and highly suspicious sequence of events in the system. Through systematic examination of the Windows Event log records (Security, System, Application), a coherent picture of a targeted anti-forensics activity emerged.